

PSC Dark I-V Analysis Project

This project focuses on analyzing the electrical behavior of a perovskite solar cell using SCAPS-1D simulation data.

Dataset Description:

The dataset (IV_doping.iv) contains voltage vs current density data along with recombination components such as SRH, Radiative, and Auger currents. The data represents both dark and illuminated conditions of the solar cell.

Objective:

To extract meaningful physical insights from the raw simulation output and visualize device behavior using scientific plots.

Graphs Generated:

1. Dark I-V Curve: Shows diode behavior of the solar cell under no illumination.
2. Semi-log I-V Curve: Highlights exponential current behavior and helps analyze diode characteristics.
3. Current Components: Displays total, recombination, and generation currents under illumination.
4. Recombination Mechanisms: Shows contribution of SRH, Radiative, and Auger recombination.

Key Observations:

- Dark current is very small and increases exponentially with voltage.
- Under illumination, photocurrent dominates and becomes negative.
- SRH recombination is the dominant loss mechanism.
- Radiative and Auger recombination are negligible in comparison.

Conclusion:

This analysis helps in understanding device physics such as charge transport, recombination losses, and overall performance behavior of perovskite solar cells. These insights are crucial for improving efficiency and optimizing device design.